

Climate Change Governance: Science, Data & Models



Lecture 1: Introduction

ENGG562/ENV562 – Fall 2026

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What is the greatest risk to humanity over the next two years?

Short term (2 years)



What is the greatest risk to humanity over the next ten years?

Long term (10 years)

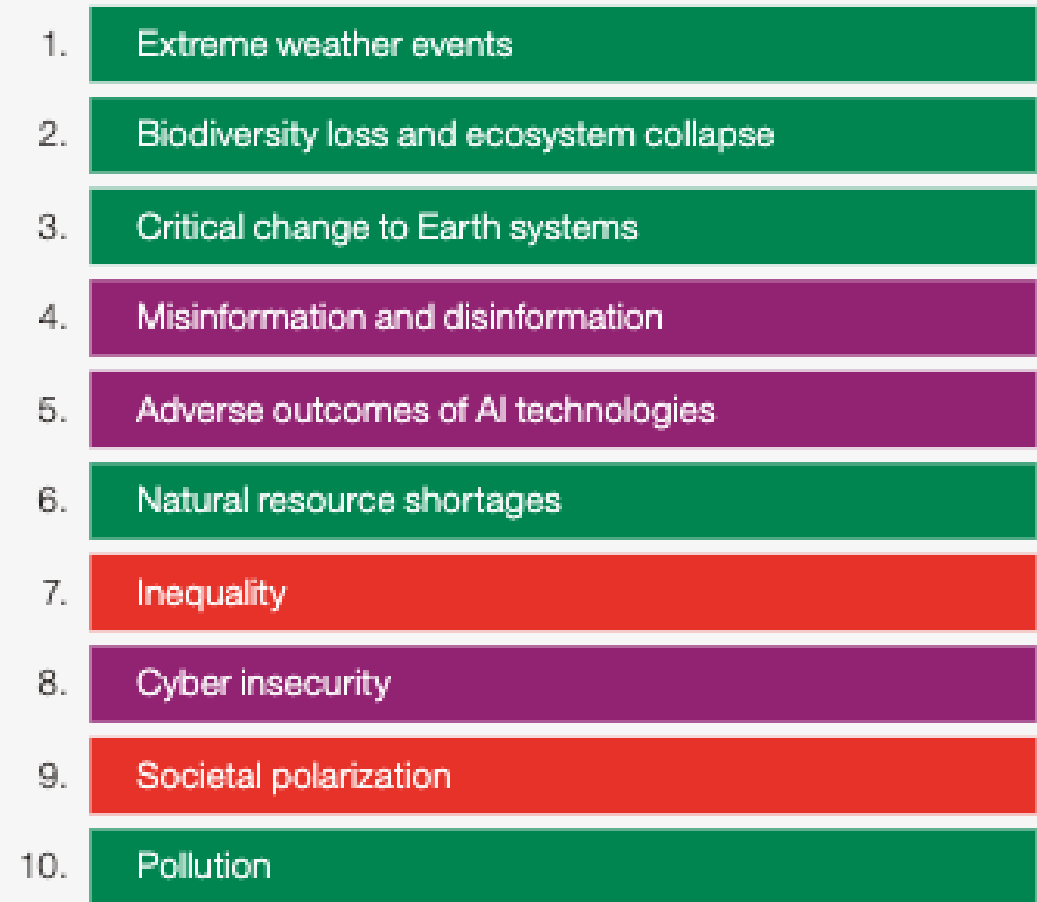


FIGURE 3

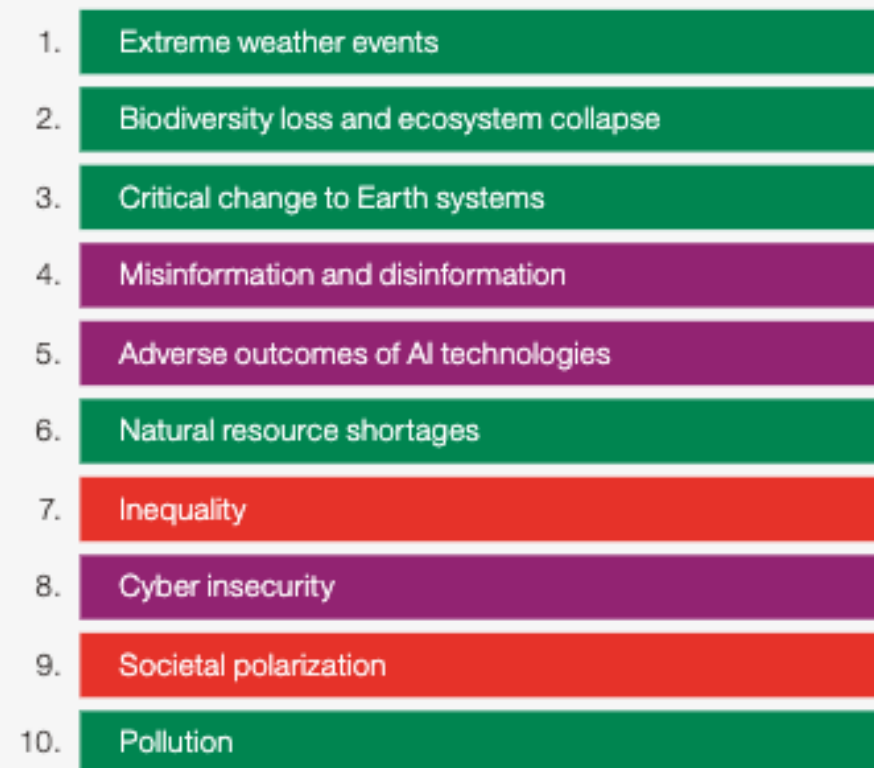
Global risks ranked by severity, short term (2 years) and long term (10 years)

"Please estimate the likely impact (severity) of the following risks over a 2-year and 10-year period."

Short term (2 years)



Long term (10 years)



Source

World Economic Forum Global Risks Perception Survey
2025-2026

Risk categories



Economic



Environmental



Geopolitical



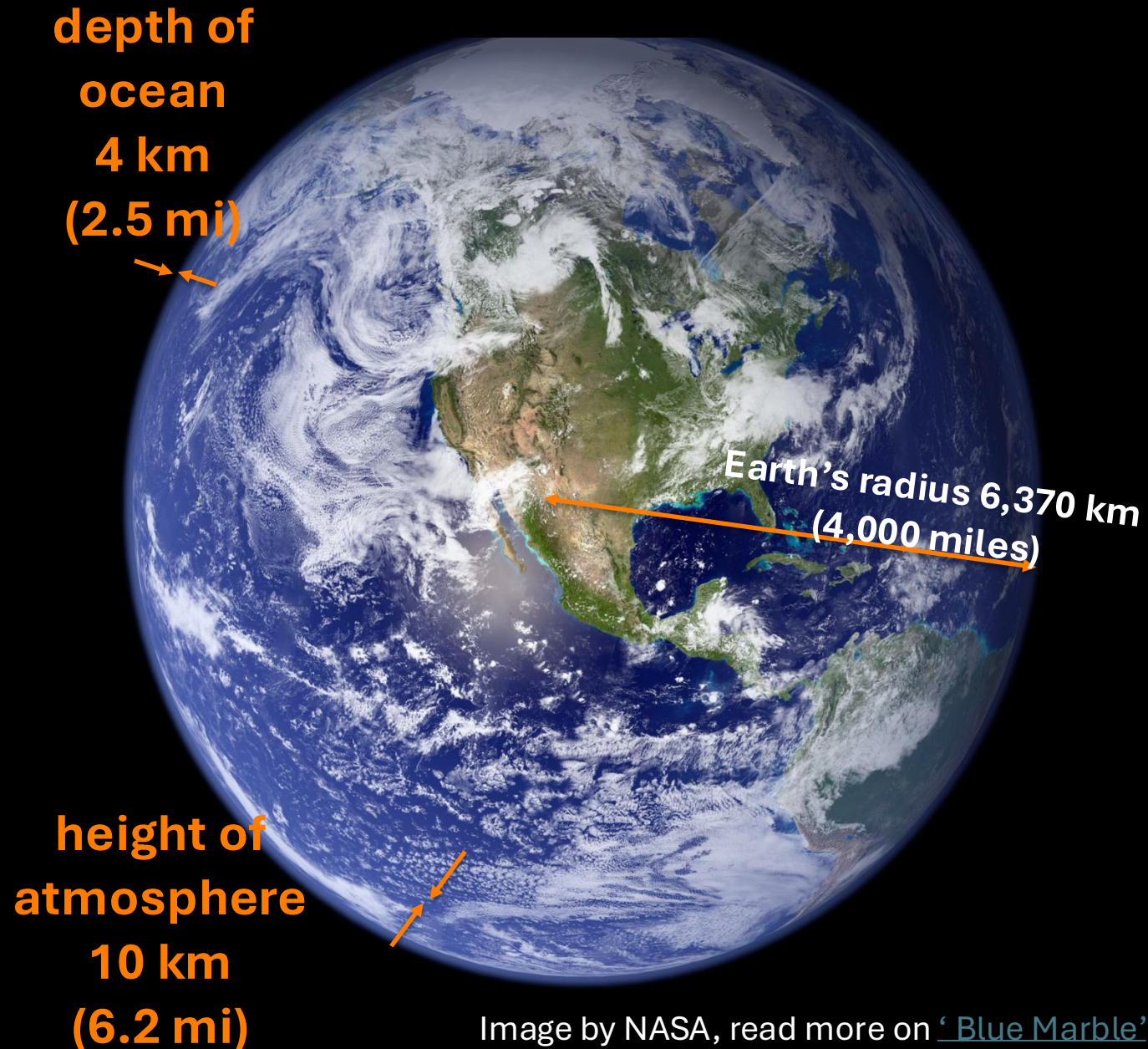
Societal



Technological

Let's talk about
Earth System

Components of Earth's Climate System



1. Atmosphere
2. Ocean
3. Cryosphere (ice & snow)
4. Biosphere

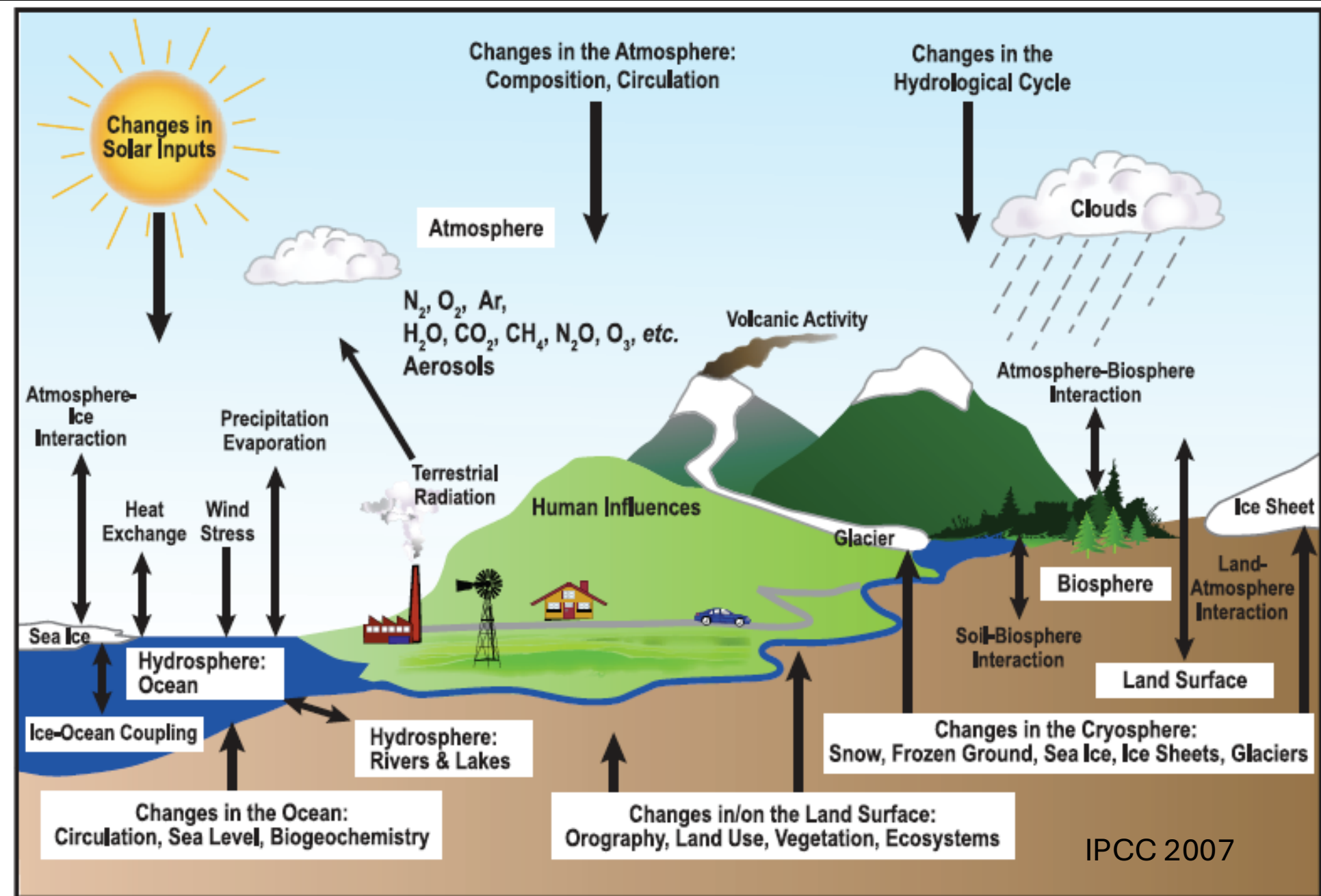
Image by NASA, read more on ['Blue Marble'](#)

Processes in the Climate System

- **Boxes** store atmosphere, ocean, land, ice, biosphere
- **Arrows** exchange

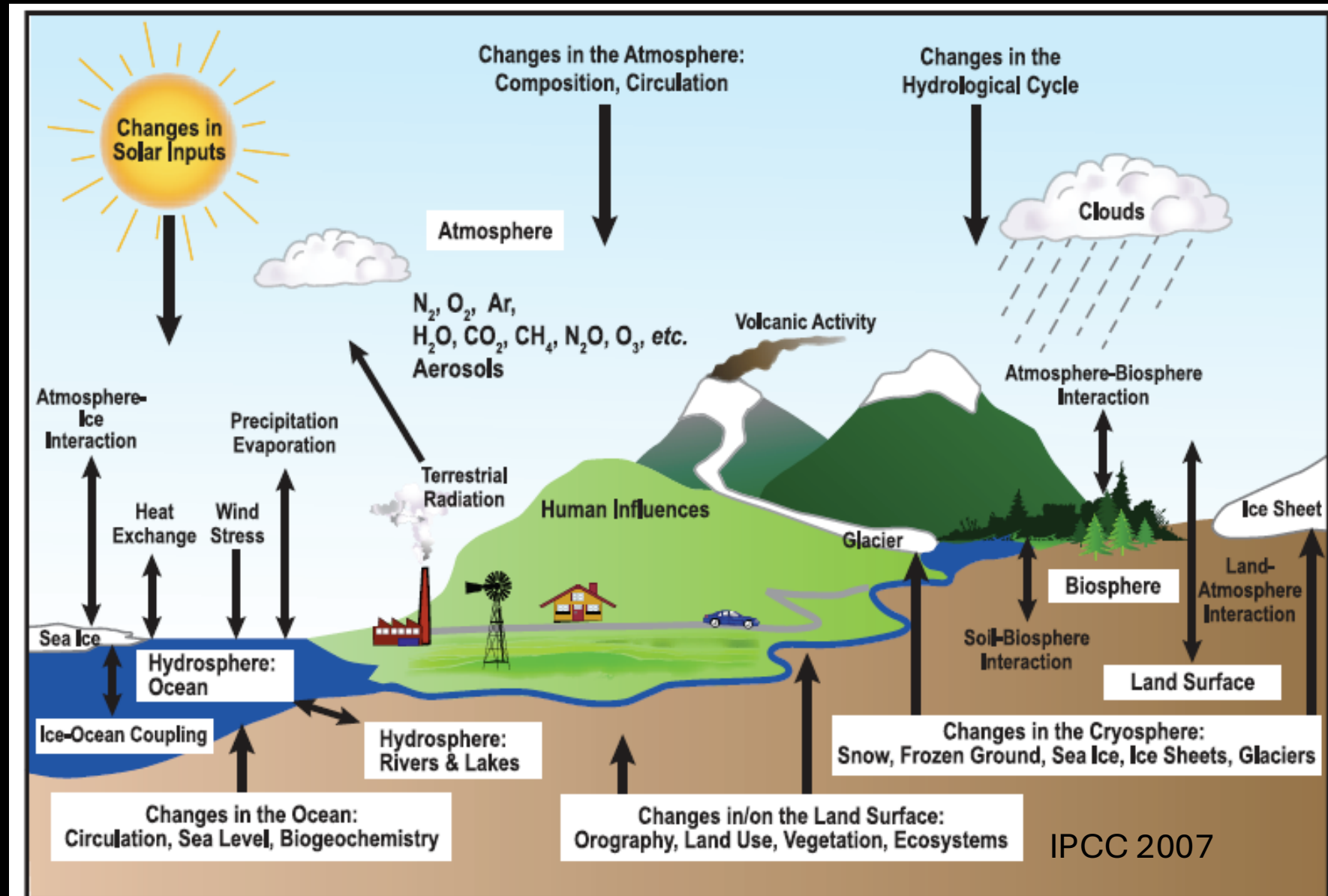
Every arrow carries one of four things:

- **energy**
- **water**
- **momentum**
- **carbon**



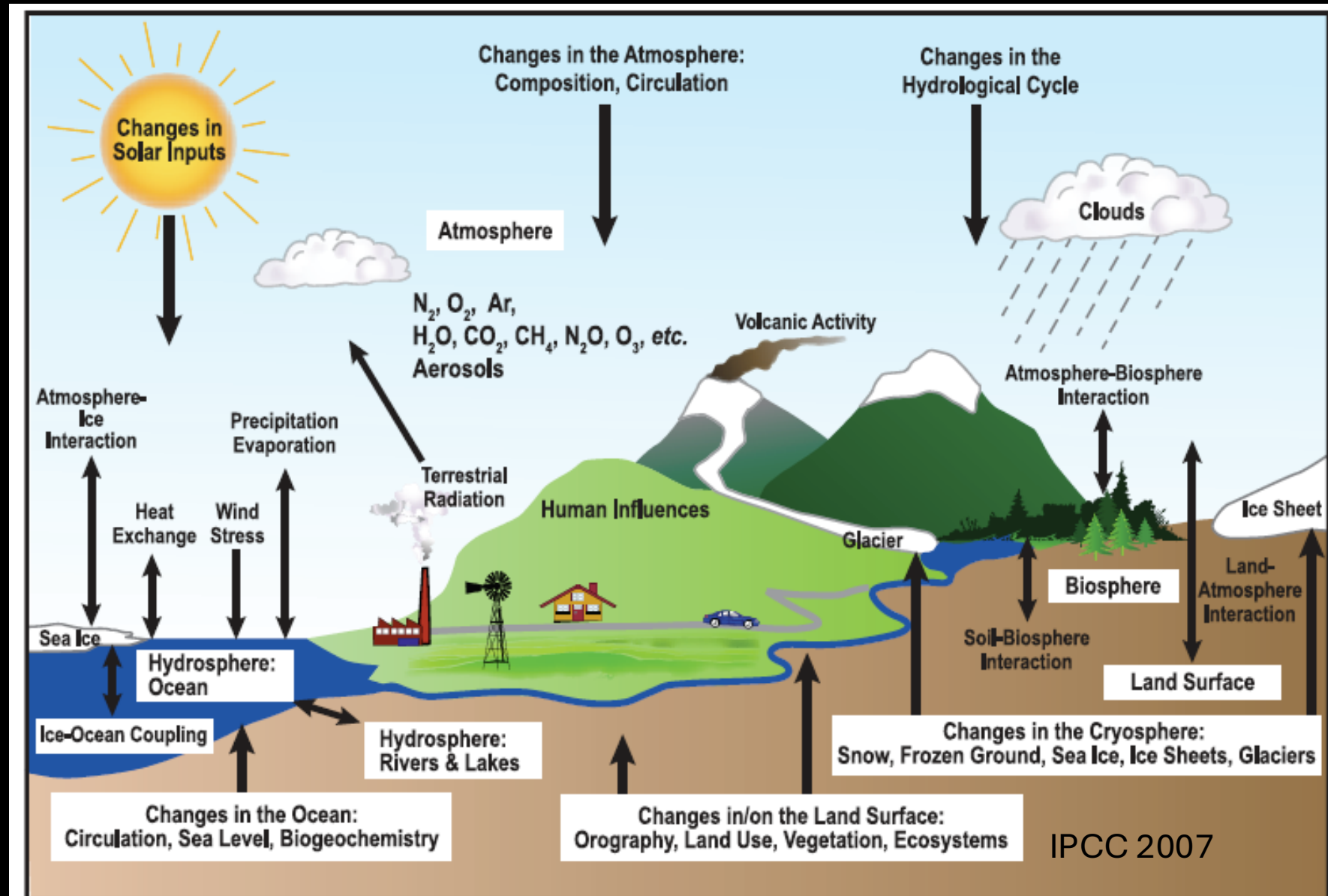
from an energy lens

- Sun in → surface warms → radiates back out as infrared
- Atmosphere intercepts part of the outgoing side → **surface ends up warmer than sunlight alone would make it**
- equator gains and poles lose



... where do humans come in ?

- Carbon: ocean ↔ air, plants ↔ air — two-way, in balance for ten thousand years



“Climate is what
we expect.
Weather is what
you get.”

WEATHER

Tells you what to wear each day



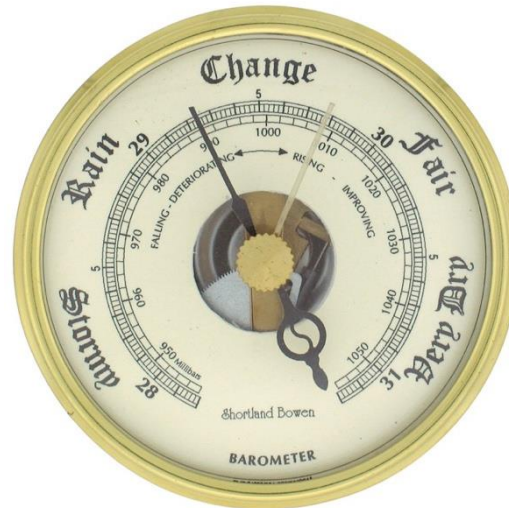
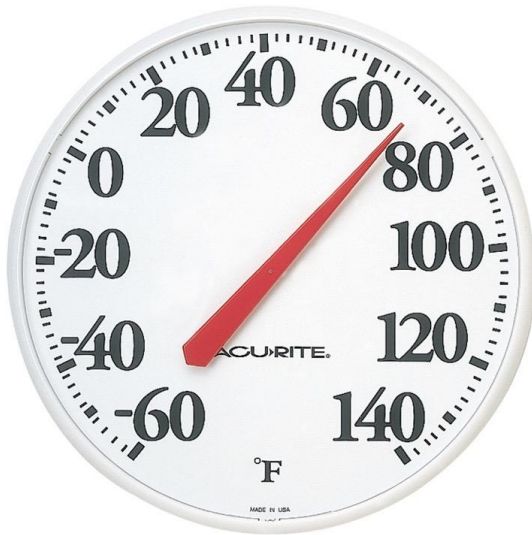
CLIMATE

Tells you what types of clothes to have in your closet



... weather is defined as

The instantaneous state of the atmosphere. Weather consists of the short-term (minutes to days) variations in the atmosphere of e.g. temperature, humidity, precipitation, cloudiness, visibility, wind. Predictability limited to days due to the non-linear, chaotic nature of the governing equations.



... climate is defined as

The average of weather. Slowly varying aspects of the atmosphere-ocean-land surface system averaged over periods of years, decades, or longer, including the variability in time of these averaged quantities and other statistics such as the probability of extreme events. Potentially predictable on long time scales (decades-centuries) and large spatial scales if forcing is known.

Summary

Earth's climate system components: atmosphere, ocean, cryosphere, biosphere

Climate is the statistics of weather, such as the long term average or the frequency of extreme events

Minute Paper (for 1-2 minute only)

- Things you expect to learn from the course
- Things from today's lecture you are still unsure about?